

Youssef Kabbaj

MSc Applied Physics — machine learning for molecular and physical systems

Route de Praz-Véguey 16, 1022 Chavannes-près-Renens, Switzerland
+41 78 817 92 20 · Born 26.06.2002
github.com/yk985 · linkedin.com/in/youssef-kabbaj

youssef.kabbaj@epfl.ch
youssefkabbaj026@gmail.com
EPFL graduate: facilitated Swiss work authorisation
Available from April 2027

Profile

MSc student in Applied Physics at EPFL (graduating March 2027), working at the interface of **statistical physics** and **generative machine learning**. Currently developing **discrete diffusion models** for conditioned **protein sequence generation**; previously built transformer-parametrised **Potts models** (attention DCA) for protein families, and a deep-learning pipeline for ultrasound motion estimation and object localization in industry. Comfortable across the stack: stochastic processes and sampling theory, PyTorch training on GPU, and molecular simulation (GROMACS, CP2K, QM/MM).

Education

EPFL — Master of Science in Applied Physics GPA 5.68 / 6 — expected March 2027	09/2024 – present
EPFL — Bachelor of Science in Physics GPA 5.69 / 6	09/2021 – 07/2024
EPFL — Cours de Mathématiques Spéciales (CMS) GPA 5.87 / 6	09/2020 – 07/2021
Groupe Scolaire Atlas, Morocco — Baccalauréat, Mathematical Sciences 18.94 / 20	07/2020

Technical skills

- **Statistical physics & stochastic modelling** — Langevin and Fokker–Planck equations, stochastic differential equations, Monte Carlo and MCMC sampling, maximum-entropy and Potts models, free-energy landscapes.
- **Generative modelling** — diffusion and score-based models, flow matching, autoregressive models, attention and transformers, energy-based models; training by maximum and pseudo-likelihood.
- **Machine learning** — supervised and reinforcement learning, synthetic data generation with ground truth, metric design for unlabelled data, model evaluation and failure analysis.
- **Molecular simulation** — set up and ran QM/MM and molecular-dynamics workflows in GROMACS, CP2K and MiMiC during a semester project; structure visualisation in PyMOL.
- **Chemistry foundations** — bond types, intermolecular interactions, binding and affinity energies: enough to work alongside chemists and read structural data critically.
- **Programming** — Python (PyTorch, NumPy, scikit-learn), C++ , MATLAB, Git, Linux; basic experience running jobs on HPC clusters.

Research and industry experience

Laboratory of Statistical Biophysics, EPFL 11/2026 – 03/2027
Master thesis (from November 2026): Diffusion models for conditioned protein sequence generation and their dynamics

- Implement a discrete diffusion model for protein sequence generation.
- Study the dynamics of generation: speciation time and mode collapse.
- Condition generation on protein family and function.

Bracco Suisse S.A. 02/2026 – 08/2026
Industry internship: deep-learning pipeline for contrast-enhanced ultrasound image processing

- Built and trained two deep-learning models on ultrasound sequences: one for **optical flow estimation**, one for **localisation**.
- Wrote a dedicated synthetic data generator for each model, designed to reproduce the appearance and statistics of real acquisitions, so that both could be measured against known ground truth.
- On synthetic data, reached a motion-correction error of roughly 5% with precision and accuracy both above 95%. No in-house baseline existed, so the evaluation framework had to be built alongside the models.

- Designed reference-free metrics to quantify performance on real, unlabelled clinical data where no ground truth exists, and fed the observed failure modes back into the data generators.

Laboratory of Computational Chemistry and Biochemistry, EPFL

09/2025 – 11/2025

Semester project: QM/MM simulation of acetone — GROMACS, CP2K, MiMiC

- Ran QM/MM simulations of acetone in MiMiC and measured how different placements of the boundary between the quantum and classical regions change the computed energies and vibrational frequencies.

Laboratory of Statistical Biophysics, EPFL

02/2025 – 07/2025

Summer internship and semester project: Biased generative protein models based on attention DCA

- Fitted a Potts model to the bacterial J-domain alignment ($L = 63$ sites, $q = 21$ states, 36,713 sequences), with the couplings parametrised by a factored self-attention layer and trained by reweighted pseudo-likelihood maximisation.
- Sampled new sequences with a Gibbs-like conditional sampler at controlled inverse temperature: generated sequences reproduce the natural one-site frequencies at $r = 0.99$ and the connected two-site correlations at $r = 0.87$, against a train-versus-test ceiling of 0.94.
- Showed the samples are genuinely novel and as diverse as natural ones (mean Hamming distance 0.42 to the nearest training sequence; mean pairwise distance 0.68 versus 0.60), and diagnosed the one clear defect: a sub-family carrying 14.1% of the natural weight but only 5.1% of the samples.
- Found that the supplied train/test files shared 79% of their sequences verbatim; rebuilt a sequence-disjoint split by clustering at 80% identity and re-evaluated everything on it. Code: github.com/yk985/attention_dca_jdomains

Swiss Plasma Center, EPFL

07/2024 – 01/2025

Semester project: Investigation of the Alfvén wave continuum in tokamaks; preceded by a Summer-in-the-Lab research internship

- Computed the Alfvén continuum spectrum of tokamak equilibria and analysed the frequency gaps opened in it by the coupling between shear Alfvén and acoustic waves.

EPFL

09/2022 – present

Teaching assistant in analysis and electromagnetism (12 h/week)

Selected personal projects

Code and documentation: github.com/yk985

- **Generative models for protein sequences** — built a hierarchical Potts model as a synthetic ground truth with known statistics, then benchmarked autoregressive and diffusion generators against the exact target distribution. github.com/yk985/hierarchical-Potts_models_learning
- **Drift-field inference from stochastic trajectories** — recovering force fields from simulated overdamped walkers, comparing five approaches from direct estimation to an amortised CNN that infers the field in a single forward pass. github.com/yk985/drift_field_inference
- **Reinforcement learning for microswimmer navigation** — RL policies for a swimmer steering through a noisy low-Reynolds-number environment, benchmarked against the exact dynamic-programming optimum so that the policy gap is measured rather than assumed. github.com/yk985/microswimmer-rl

Languages

Arabic — mother tongue · French — C1 · English — C1

Willing to learn German for a role in German-speaking Switzerland.

Interests

Basketball and sport in general; contemplation and the larger questions about life. I will bring home-made tiramisu to the office.